

Setting up the rust investigation

One sheet per group. These jars sit on the shelf for five weeks — you do not get to start again.

Group	
Our condition	
Date set up	

Our prediction

How much rust will your condition make compared with everyone else's — most in the room, least, or somewhere in the middle? Say why.

Steps

1. **Check your jars.** Any chips or cracks, swap it.
2. **Mark each nail** halfway along with the marker. That line is what “half covered” means — not the jar.
3. **Weigh 3 nails together.** Write the number in the table below. Do this for each jar.
4. **Put 3 nails in each jar.**
5. **Set up your condition** in all 3 jars, the same way every time. Half covered = water to the marker line. Fully covered = at least 1 cm over the nail.
6. **Draw a line on the outside** of each jar where the water comes to. You top it back up to that line every week.
7. **Fill in and cut off the labels** at the bottom. Stick one on each jar.
8. **Lid on or off?** Only Tuesday A and F get a lid. Everything else stays open.
9. **Write the repeat sentence.** Then put the jars on the shelf.

Why three jars and not one If you only had one and something went wrong with it, you would never know. Three lets you spot the odd one out.

Starting mass

	Jar 1	Jar 2	Jar 3
Mass of 3 nails (g)			

Record to the precision your scales actually give you. If the scales read whole grams, do not write decimal places — that is an error, not extra care.

The repeat sentence

Write the ONE sentence that would let another group set this up exactly the same way, without asking you anything. Name the nails, the liquid, how much, and whether the lid is on.

Check it against this: “Three 65 mm nails, half covered in tap water, lid off, on the shelf by the window.” Does yours give that much?

Cut off these labels and stick one on each jar

RUST INVESTIGATION Group: _____ Condition: _____ Jar size: _____ Jar 1 of 3 Nail mass: _____ g Date: _____	RUST INVESTIGATION Group: _____ Condition: _____ Jar size: _____ Jar 2 of 3 Nail mass: _____ g Date: _____	RUST INVESTIGATION Group: _____ Condition: _____ Jar size: _____ Jar 3 of 3 Nail mass: _____ g Date: _____
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Jar G group You have a fourth jar. Write your own label for it: Jar 4 of 4, condition G, and the mass of the three ordinary nails only.

Safety Glasses on. Nails counted out and counted back in. Jars stay on the bench and are carried with two hands, one at a time. Report any spill straight away and clean it up.